Evaluating Prediction Intervals

Study Using the M3 Competition Dataset

Igor Faluhelyi

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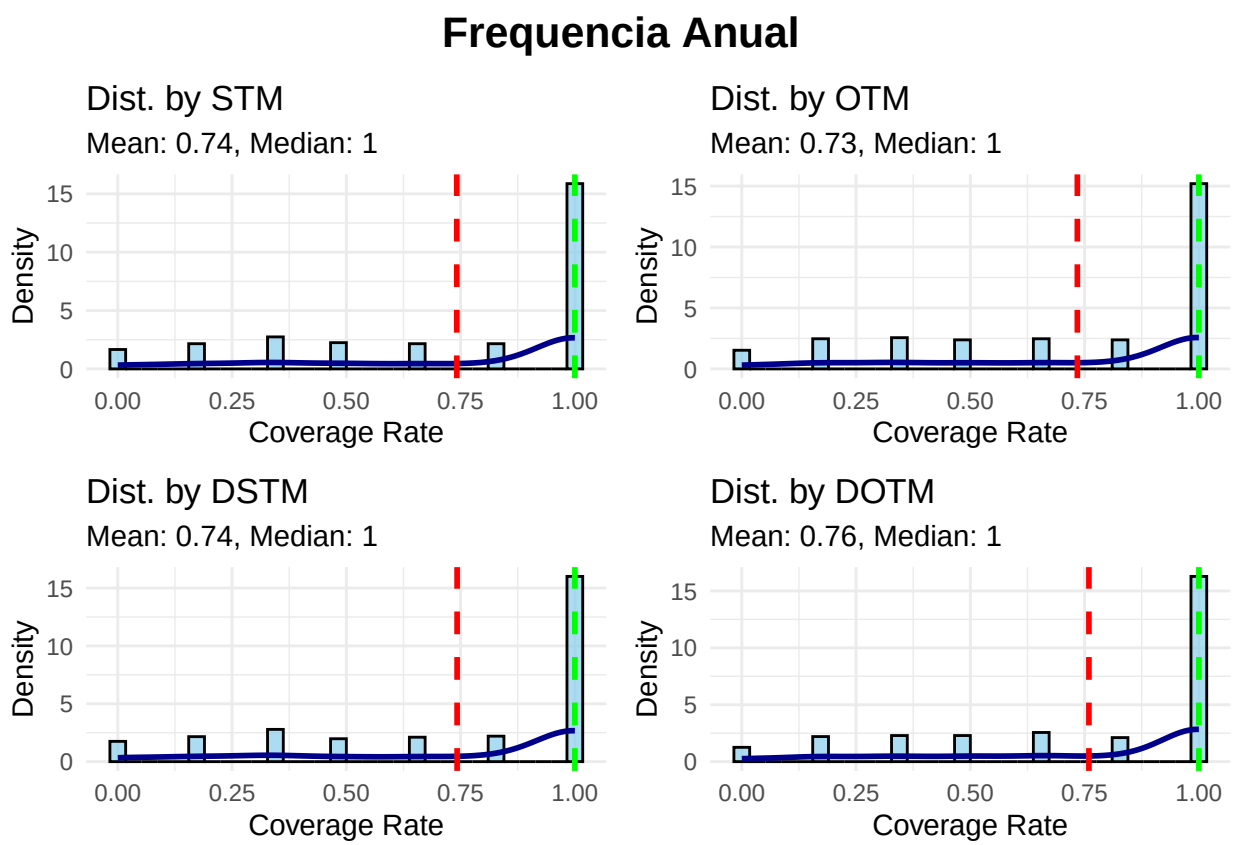
1 Avaliação da taxa de cobertura

1.1 Frequência Anual

```r
# Process the model type and calculate coverage rates
process_model <- function(model_key, label, M3, results, n_rows = 3003, n_cols = 18) {
  lo <- matrix(NA, nrow = n_rows, ncol = n_cols)
  hi <- matrix(NA, nrow = n_rows, ncol = n_cols)
  obs <- matrix(NA, nrow = n_rows, ncol = n_cols)
  coverage_rates <- c()
  c <- 1

  for (i in 1:n_rows) {
    x <- M3[[i]]$x
    h <- M3[[i]]$h
    if (M3[[i]]$period == "YEARLY") {
      obs[i, 1:h] <- M3[[i]]$xx
      lo[i, 1:h] <- results[[model_key]][[c]]$lower[, 3] # 95% interval
      hi[i, 1:h] <- results[[model_key]][[c]]$upper[, 3]
      c <- c + 1
```

```r
      coverage_rates <- c(
        coverage_rates,
        calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
      )
    }
  }

  evaluate_coverage_rates(coverage_rates, label)
}

# Process each model and store the results
results_list <- list(
  STM = process_model("stm_anua", "STM", M3, results),
  OTM = process_model("otm_anua", "OTM", M3, results),
  DSTM = process_model("dstm_anua", "DSTM", M3, results),
  DOTM = process_model("dotm_anua", "DOTM", M3, results)
)

# Plot results
titulo_principal <- textGrob("Frequencia Anual",
                             gp = gpar(fontsize = 16, fontface = "bold"))

grid.arrange(
  titulo_principal,
  arrangeGrob(
    results_list$STM$plot,
    results_list$OTM$plot,
    results_list$DSTM$plot,
    results_list$DOTM$plot,
    nrow = 2
  ),
  nrow = 2,
  heights = c(0.1, 1)
)
```

```
## Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(density)` instead.
## i The deprecated feature was likely used in the forecTheta package.
##   Please report the issue at <https://github.com/jafiorucci/forecTheta/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

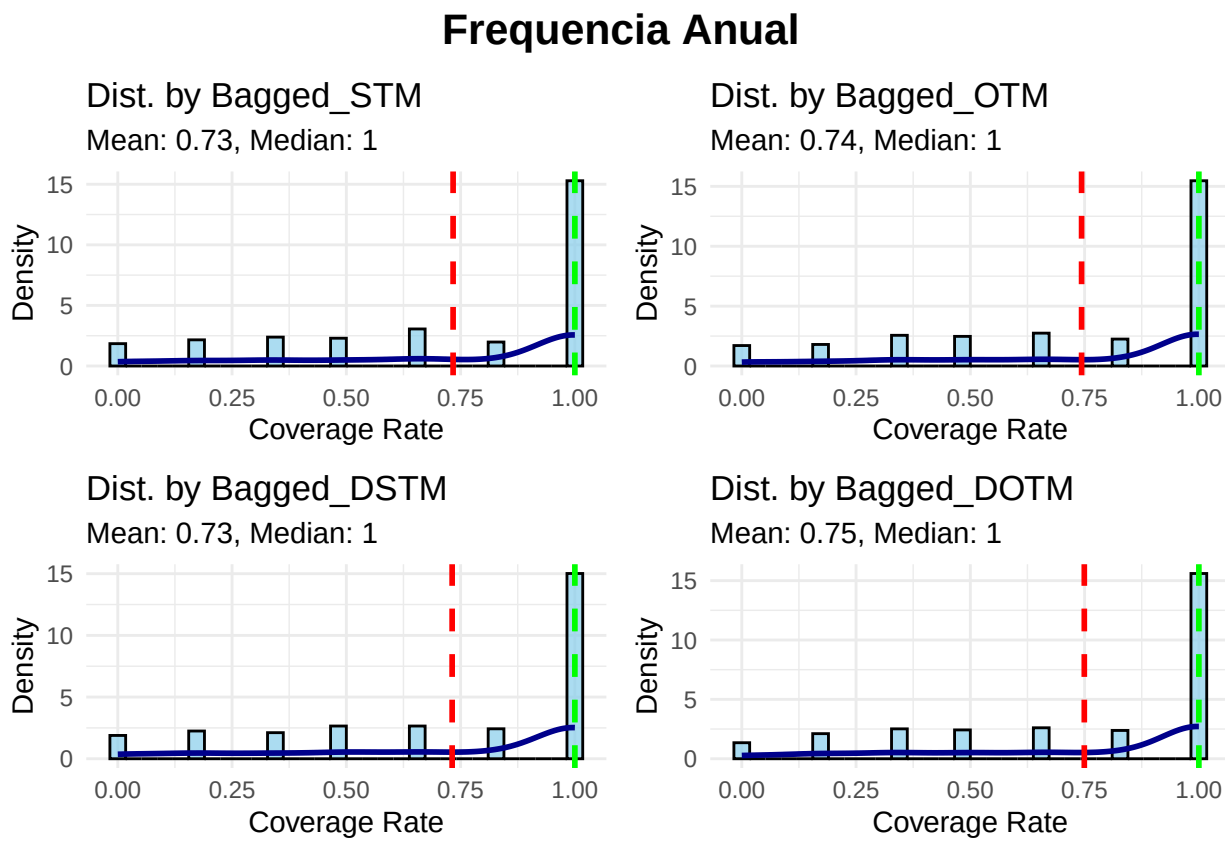



```r
# Combine summaries for comparison
summaries <- cbind(
  DOTM = results_list$DOTM$summary,
  DSTM = results_list$DSTM$summary,
  OTM = results_list$OTM$summary,
  STM = results_list$STM$summary
)
summaries
```

```
##         DOTM      DSTM      OTM       STM
## mean    0.7594315 0.7423773 0.7341085 0.7416021
## median  1         1         1         1
## sd      0.3243606 0.3404963 0.3361287 0.3379349
## min     0         0         0         0
## max     1         1         1         1
```

```r
results_list <- list(
  Bagged_STM = process_model("bagged_stm_anua", "Bagged_STM", M3, results),
  Bagged_OTM = process_model("bagged_otm_anua", "Bagged_OTM", M3, results),
  Bagged_DSTM = process_model("bagged_dstm_anua", "Bagged_DSTM", M3, results),
  Bagged_DOTM = process_model("bagged_dotm_anua", "Bagged_DOTM", M3, results)
)

titulo_principal <- textGrob("Frequencia Anual",
                             gp = gpar(fontsize = 16, fontface = "bold"))

grid.arrange(
```

```r
  titulo_principal,
  arrangeGrob(
    results_list$Bagged_STM$plot,
    results_list$Bagged_OTM$plot,
    results_list$Bagged_DSTM$plot,
    results_list$Bagged_DOTM$plot,
    nrow = 2
  ),
  nrow = 2,
  heights = c(0.1, 1)
)
```

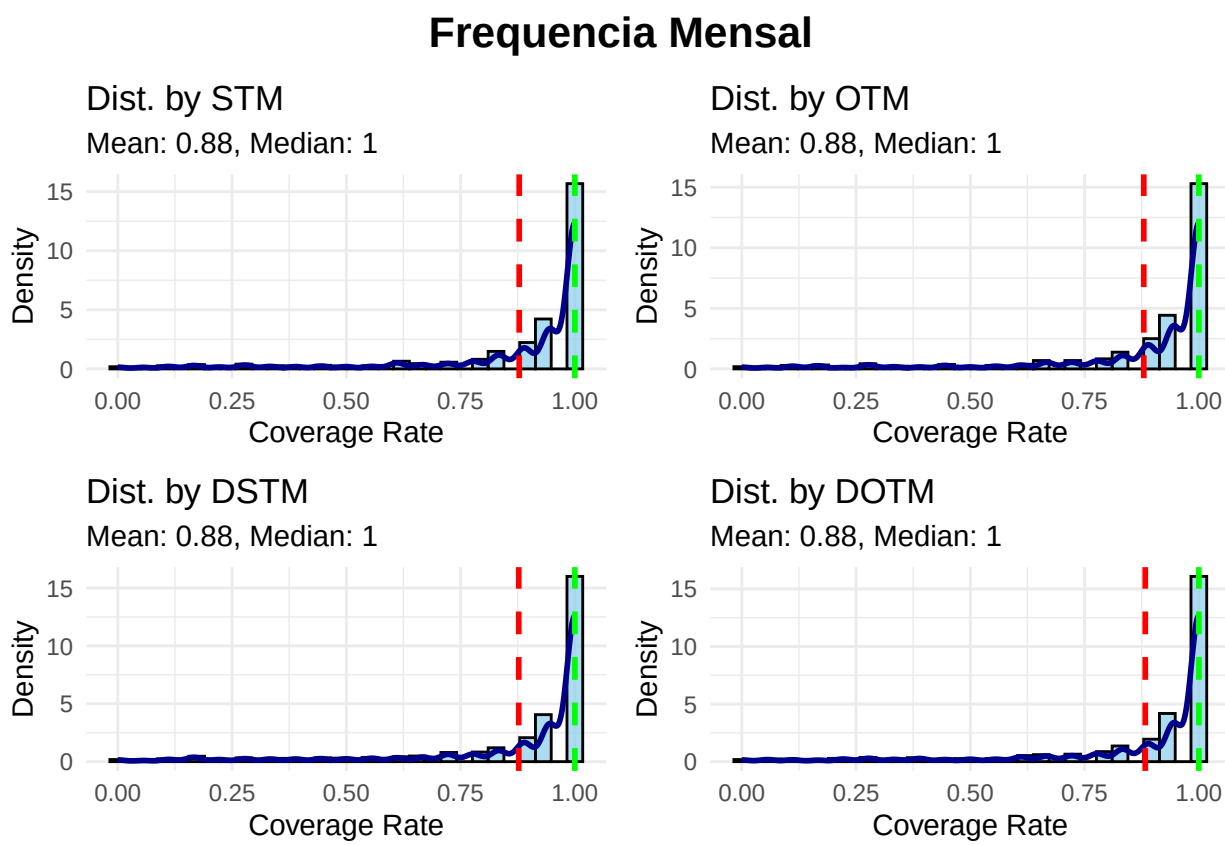



```r
summaries <- cbind(
  DOTM = results_list$Bagged_DOTM$summary,
  DSTM = results_list$Bagged_DSTM$summary,
  OTM = results_list$Bagged_OTM$summary,
  STM = results_list$Bagged_STM$summary
)
summaries
```

```
##         DOTM      DSTM      OTM       STM
## mean    0.749354  0.7315245 0.7434109 0.7335917
## median  1         1         1         1
## sd      0.3259562 0.3388088 0.3310117 0.3381509
## min     0         0         0         0
## max     1         1         1         1
```

1.2 Frequência Mensal

```r
# Process the model type and calculate coverage rates
process_model <- function(model_key, label, M3, results, n_rows = 3003, n_cols = 18) {
  lo <- matrix(NA, nrow = n_rows, ncol = n_cols)
  hi <- matrix(NA, nrow = n_rows, ncol = n_cols)
  obs <- matrix(NA, nrow = n_rows, ncol = n_cols)
  coverage_rates <- c()
  c <- 1

  for (i in 1:n_rows) {
    x <- M3[[i]]$x
    h <- M3[[i]]$h
    if (M3[[i]]$period == "MONTHLY") {
      obs[i, 1:h] <- M3[[i]]$xx
      lo[i, 1:h] <- results[[model_key]][[c]]$lower[, 3] # 95% interval
      hi[i, 1:h] <- results[[model_key]][[c]]$upper[, 3]
      c <- c + 1
      coverage_rates <- c(
        coverage_rates,
        calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
      )
    }
  }

  evaluate_coverage_rates(coverage_rates, label)
}

# Process each model and store the results
results_list <- list(
  STM = process_model("stm_mens", "STM", M3, results),
  OTM = process_model("otm_mens", "OTM", M3, results),
  DSTM = process_model("dstm_mens", "DSTM", M3, results),
  DOTM = process_model("dotm_mens", "DOTM", M3, results)
)

# Plot results
titulo_principal <- textGrob("Frequencia Mensal",
                             gp = gpar(fontsize = 16, fontface = "bold"))

grid.arrange(
  titulo_principal,
  arrangeGrob(
    results_list$STM$plot,
    results_list$OTM$plot,
    results_list$DSTM$plot,
    results_list$DOTM$plot,
    nrow = 2
  ),
  nrow = 2,
  heights = c(0.1, 1)
)
```

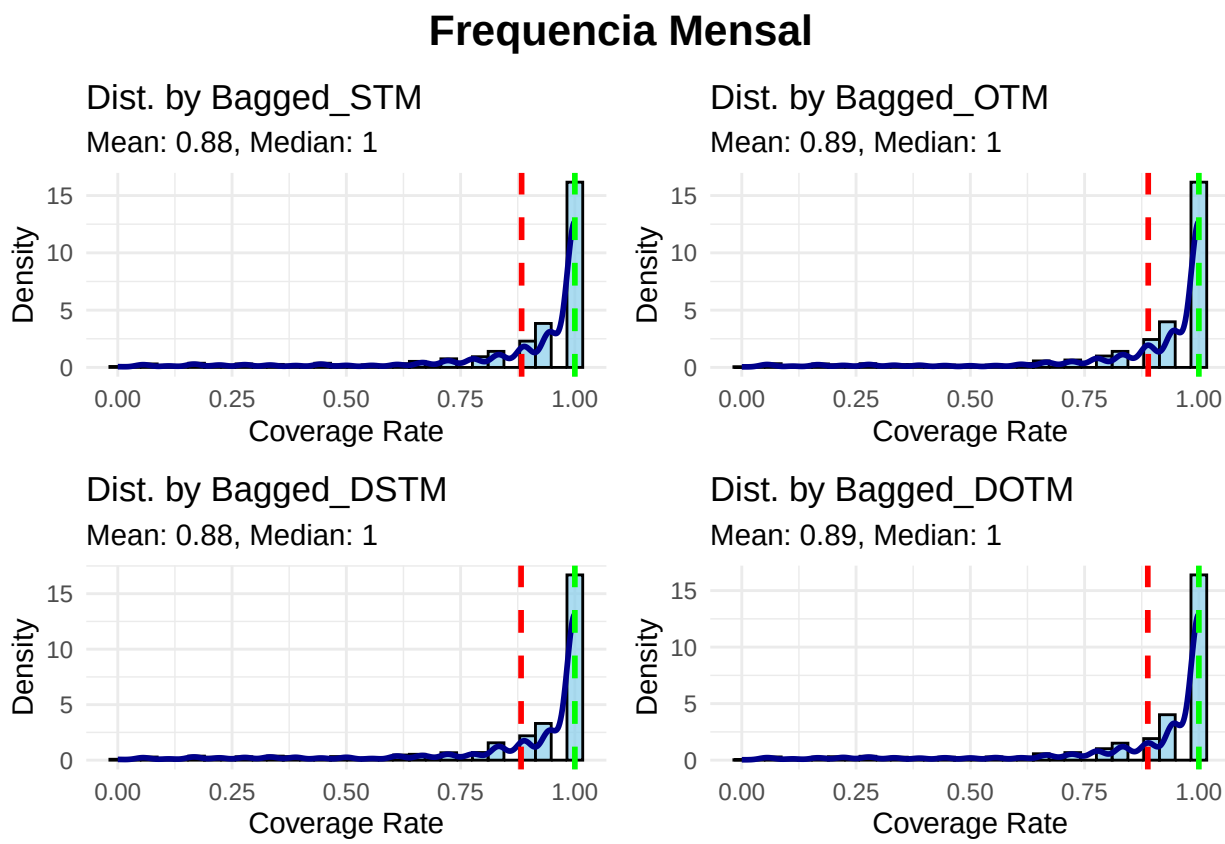



```r
# Combine summaries for comparison
summaries <- cbind(
  DOTM = results_list$DOTM$summary,
  DSTM = results_list$DSTM$summary,
  OTM = results_list$OTM$summary,
  STM = results_list$STM$summary
)
summaries
```

```
##         DOTM      DSTM      OTM       STM
## mean    0.882742  0.8772565 0.8795518 0.878268
## median  1         1         1         1
## sd      0.2165661 0.2236503 0.2176576 0.2216902
## min     0         0         0         0
## max     1         1         1         1
```

```r
results_list <- list(
  Bagged_STM = process_model("bagged_stm_mens", "Bagged_STM", M3, results),
  Bagged_OTM = process_model("bagged_otm_mens", "Bagged_OTM", M3, results),
  Bagged_DSTM = process_model("bagged_dstm_mens", "Bagged_DSTM", M3, results),
  Bagged_DOTM = process_model("bagged_dotm_mens", "Bagged_DOTM", M3, results)
)


titulo_principal <- textGrob("Frequencia Mensal",
                             gp = gpar(fontsize = 16, fontface = "bold"))
```

```r
grid.arrange(
  titulo_principal,
  arrangeGrob(
    results_list$Bagged_STM$plot,
    results_list$Bagged_OTM$plot,
    results_list$Bagged_DSTM$plot,
    results_list$Bagged_DOTM$plot,
    nrow = 2
  ),
  nrow = 2,
  heights = c(0.1, 1)
)
```



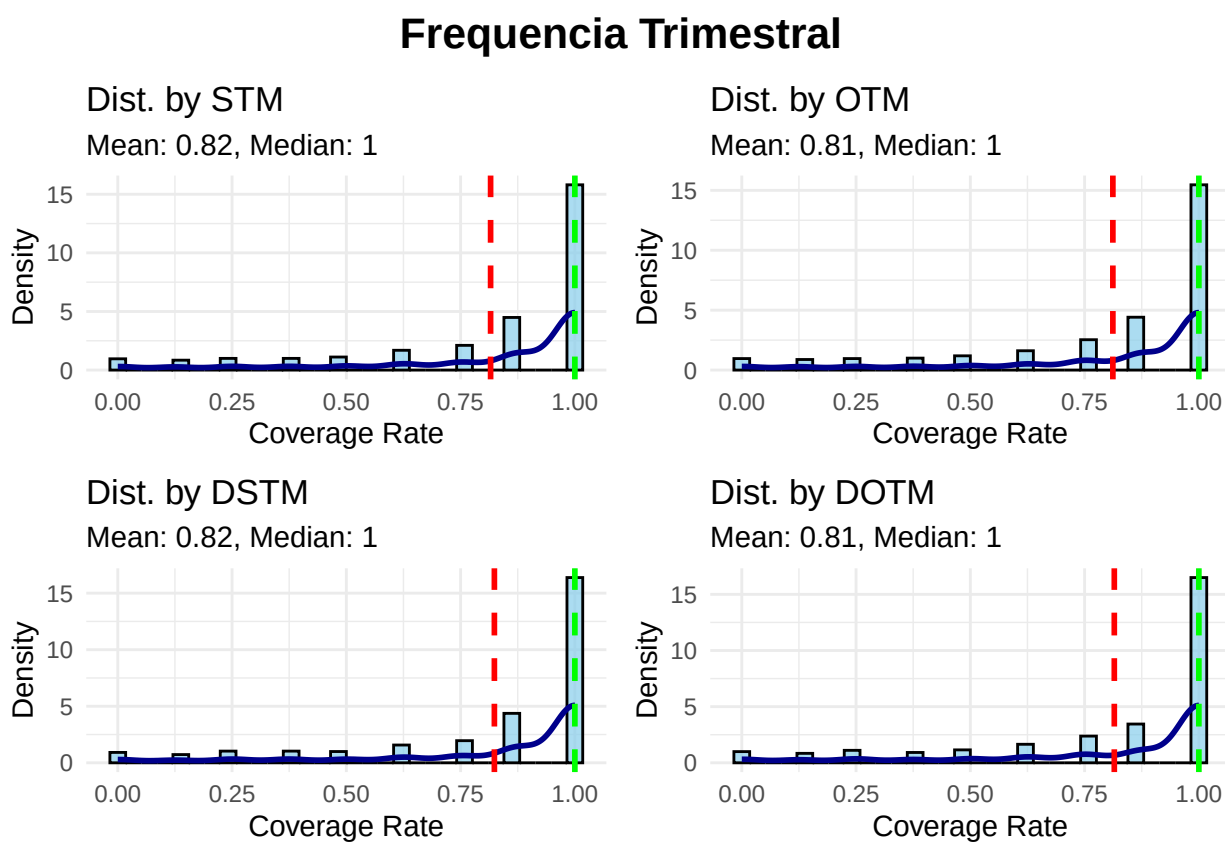

```r
summaries <- cbind(
  DOTM = results_list$Bagged_DOTM$summary,
  DSTM = results_list$Bagged_DSTM$summary,
  OTM = results_list$Bagged_OTM$summary,
  STM = results_list$Bagged_STM$summary
)
summaries
```

```
##         DOTM      DSTM      OTM       STM
## mean    0.8883053 0.882314  0.8892001 0.8835201
## median  1         1         1         1
## sd      0.2097078 0.2190214 0.2090938 0.2164169
## min     0         0         0         0
```

```
## max    1          1          1          1
```

1.3 Frequência Trimestral

```r
# Process the model type and calculate coverage rates
process_model <- function(model_key, label, M3, results, n_rows = 3003, n_cols = 18) {
  lo <- matrix(NA, nrow = n_rows, ncol = n_cols)
  hi <- matrix(NA, nrow = n_rows, ncol = n_cols)
  obs <- matrix(NA, nrow = n_rows, ncol = n_cols)
  coverage_rates <- c()
  c <- 1

  for (i in 1:n_rows) {
    x <- M3[[i]]$x
    h <- M3[[i]]$h
    if (M3[[i]]$period == "QUARTERLY") {
      obs[i, 1:h] <- M3[[i]]$xx
      lo[i, 1:h] <- results[[model_key]][[c]]$lower[, 3] # 95% interval
      hi[i, 1:h] <- results[[model_key]][[c]]$upper[, 3]
      c <- c + 1
      coverage_rates <- c(
        coverage_rates,
        calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
      )
    }
  }

  evaluate_coverage_rates(coverage_rates, label)
}

# Process each model and store the results
results_list <- list(
  STM = process_model("stm_trim", "STM", M3, results),
  OTM = process_model("otm_trim", "OTM", M3, results),
  DSTM = process_model("dstm_trim", "DSTM", M3, results),
  DOTM = process_model("dotm_trim", "DOTM", M3, results)
)

# Plot results
titulo_principal <- textGrob("Frequencia Trimestral",
                             gp = gpar(fontsize = 16, fontface = "bold"))

grid.arrange(
  titulo_principal,
  arrangeGrob(
    results_list$STM$plot,
    results_list$OTM$plot,
    results_list$DSTM$plot,
    results_list$DOTM$plot,
    nrow = 2
  ),
  nrow = 2,
  heights = c(0.1, 1)
)
```

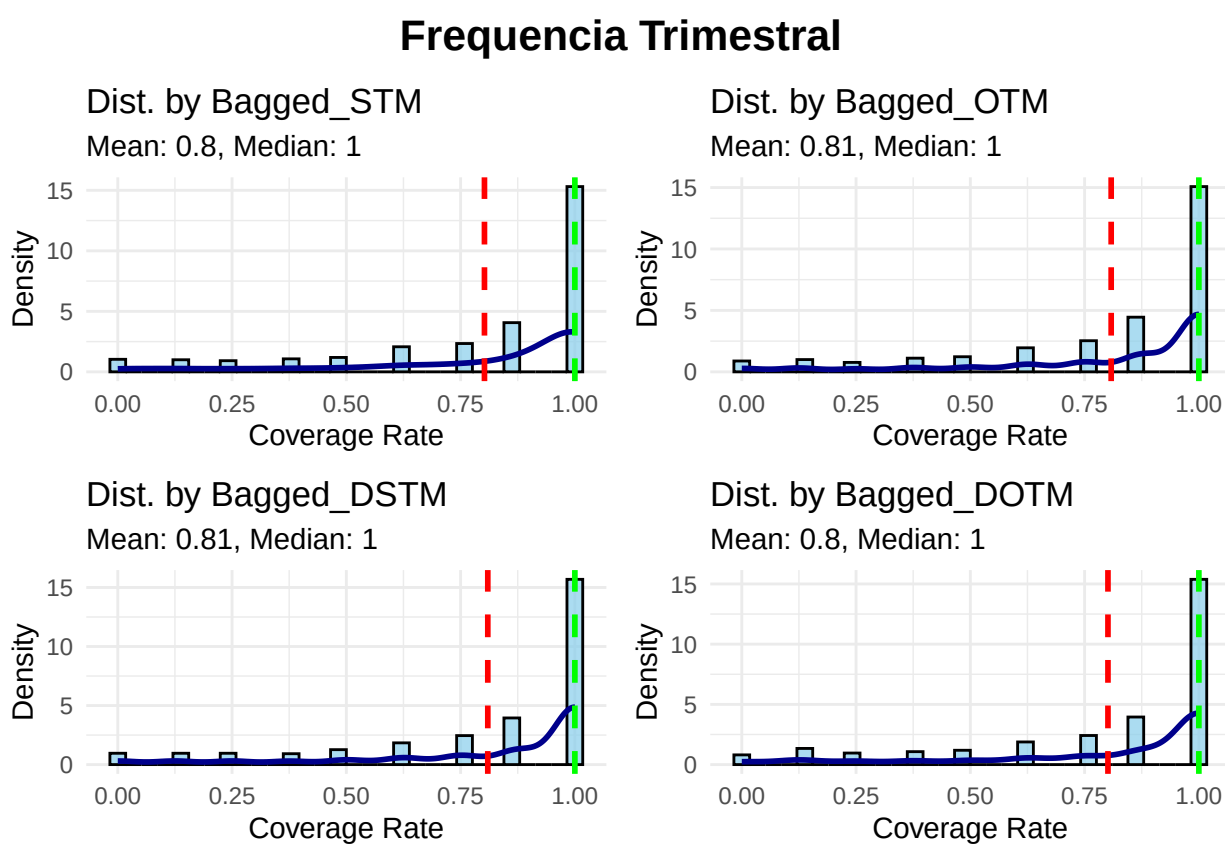



```r
# Combine summaries for comparison
summaries <- cbind(
  DOTM = results_list$DOTM$summary,
  DSTM = results_list$DSTM$summary,
  OTM = results_list$OTM$summary,
  STM = results_list$STM$summary
)
summaries
```

```
##         DOTM      DSTM      OTM       STM
## mean    0.8149802 0.8239087 0.8118386 0.8156415
## median  1         1         1         1
## sd      0.2853596 0.2773419 0.2810583 0.2810415
## min     0         0         0         0
## max     1         1         1         1
```

```r
results_list <- list(
  Bagged_STM = process_model("bagged_stm_trim", "Bagged_STM", M3, results),
  Bagged_OTM = process_model("bagged_otm_trim", "Bagged_OTM", M3, results),
  Bagged_DSTM = process_model("bagged_dstm_trim", "Bagged_DSTM", M3, results),
  Bagged_DOTM = process_model("bagged_dotm_trim", "Bagged_DOTM", M3, results)
)


titulo_principal <- textGrob("Frequencia Trimestral",
                             gp = gpar(fontsize = 16, fontface = "bold"))
```

```r
grid.arrange(
  titulo_principal,
  arrangeGrob(
    results_list$Bagged_STM$plot,
    results_list$Bagged_OTM$plot,
    results_list$Bagged_DSTM$plot,
    results_list$Bagged_DOTM$plot,
    nrow = 2
  ),
  nrow = 2,
  heights = c(0.1, 1)
)
```



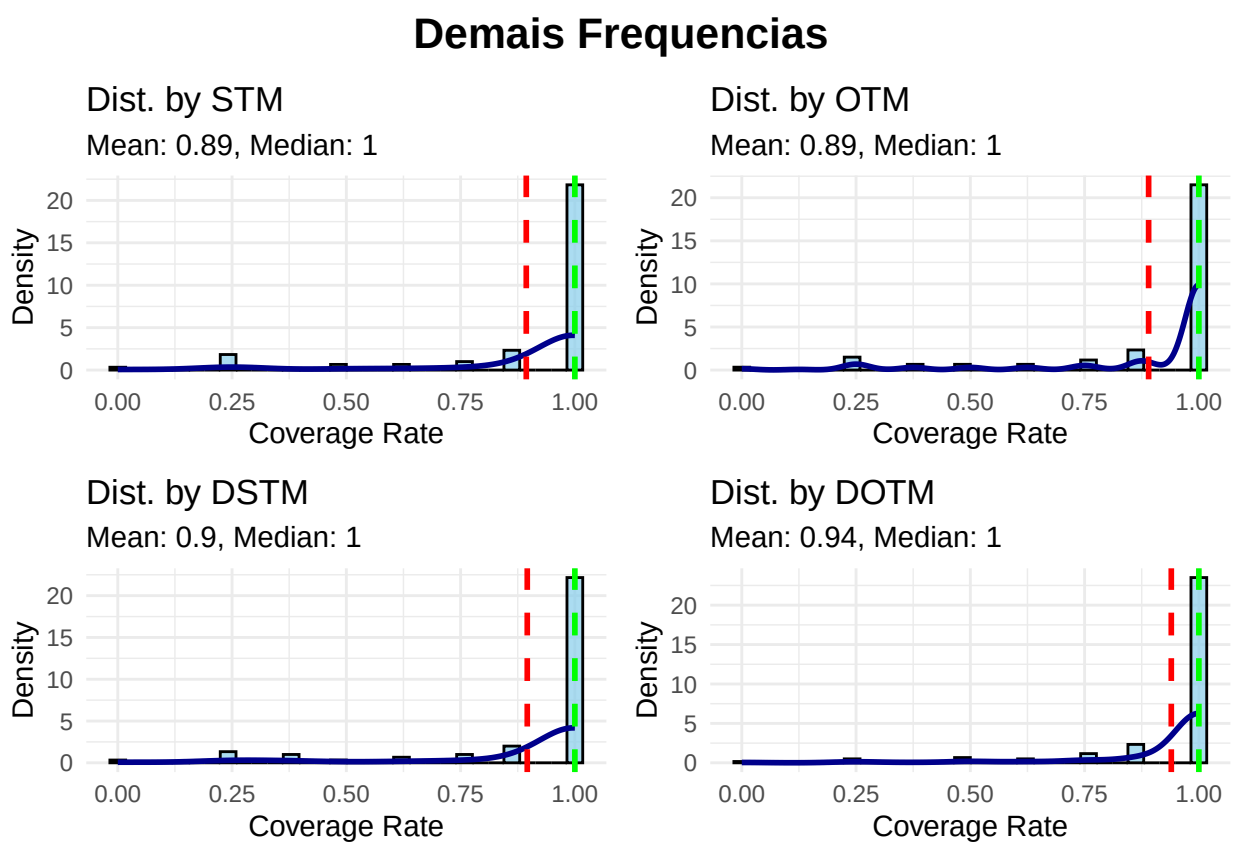

```r
summaries <- cbind(
  DOTM = results_list$Bagged_DOTM$summary,
  DSTM = results_list$Bagged_DSTM$summary,
  OTM = results_list$Bagged_OTM$summary,
  STM = results_list$Bagged_STM$summary
)
summaries
```

```
##         DOTM      DSTM      OTM       STM
## mean    0.8010913 0.8095238 0.8082011 0.8022487
## median  1         1         1         1
## sd      0.2890346 0.2836819 0.2790307 0.2879303
## min     0         0         0         0
```

```
## max    1          1          1          1
```

1.4 Demais Frequências

```r
# Process the model type and calculate coverage rates
process_model <- function(model_key, label, M3, results, n_rows = 3003, n_cols = 18) {
  lo <- matrix(NA, nrow = n_rows, ncol = n_cols)
  hi <- matrix(NA, nrow = n_rows, ncol = n_cols)
  obs <- matrix(NA, nrow = n_rows, ncol = n_cols)
  coverage_rates <- c()
  c <- 1

  for (i in 1:n_rows) {
    x <- M3[[i]]$x
    h <- M3[[i]]$h
    if (M3[[i]]$period == "OTHER") {
      obs[i, 1:h] <- M3[[i]]$xx
      lo[i, 1:h] <- results[[model_key]][[c]]$lower[, 3] # 95% interval
      hi[i, 1:h] <- results[[model_key]][[c]]$upper[, 3]
      c <- c + 1
      coverage_rates <- c(
        coverage_rates,
        calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
      )
    }
  }

  evaluate_coverage_rates(coverage_rates, label)
}

# Process each model and store the results
results_list <- list(
  STM = process_model("stm_outr", "STM", M3, results),
  OTM = process_model("otm_outr", "OTM", M3, results),
  DSTM = process_model("dstm_outr", "DSTM", M3, results),
  DOTM = process_model("dotm_outr", "DOTM", M3, results)
)

# Plot results
titulo_principal <- textGrob("Demais Frequencias",
                             gp = gpar(fontsize = 16, fontface = "bold"))

grid.arrange(
  titulo_principal,
  arrangeGrob(
    results_list$STM$plot,
    results_list$OTM$plot,
    results_list$DSTM$plot,
    results_list$DOTM$plot,
    nrow = 2
  ),
  nrow = 2,
  heights = c(0.1, 1)
)
```

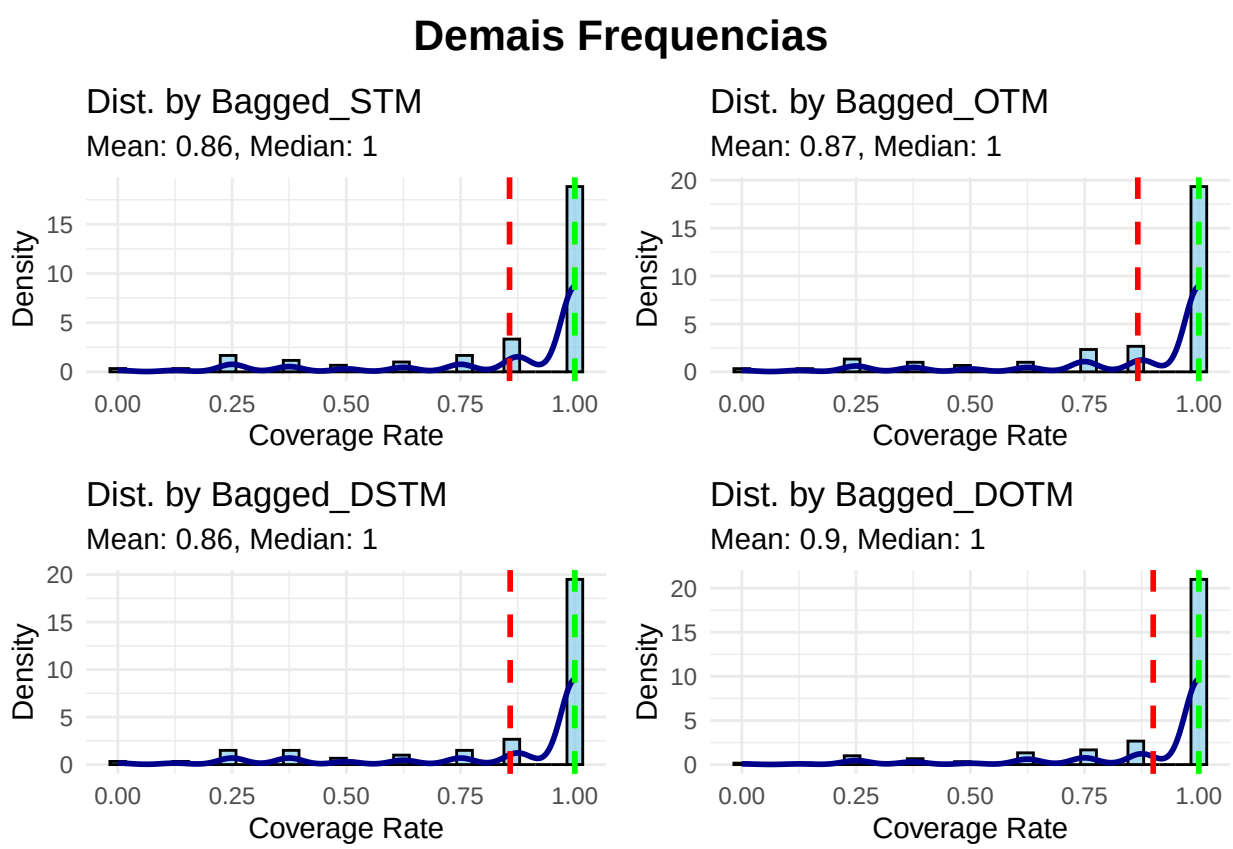



```r
# Combine summaries for comparison
summaries <- cbind(
  DOTM = results_list$DOTM$summary,
  DSTM = results_list$DSTM$summary,
  OTM = results_list$OTM$summary,
  STM = results_list$STM$summary
)
summaries
```

```
##         DOTM      DSTM      OTM       STM
## mean    0.9396552 0.8958333 0.8900862 0.8936782
## median  1         1         1         1
## sd      0.161686  0.2324331 0.2344256 0.2347431
## min     0         0         0         0
## max     1         1         1         1
```

```r
results_list <- list(
  Bagged_STM = process_model("bagged_stm_outr", "Bagged_STM", M3, results),
  Bagged_OTM = process_model("bagged_otm_outr", "Bagged_OTM", M3, results),
  Bagged_DSTM = process_model("bagged_dstm_outr", "Bagged_DSTM", M3, results),
  Bagged_DOTM = process_model("bagged_dotm_outr", "Bagged_DOTM", M3, results)
)


titulo_principal <- textGrob("Demais Frequencias",
                             gp = gpar(fontsize = 16, fontface = "bold"))
```

```r
grid.arrange(
  titulo_principal,
  arrangeGrob(
    results_list$Bagged_STM$plot,
    results_list$Bagged_OTM$plot,
    results_list$Bagged_DSTM$plot,
    results_list$Bagged_DOTM$plot,
    nrow = 2
  ),
  nrow = 2,
  heights = c(0.1, 1)
)
```




```r
summaries <- cbind(
  DOTM = results_list$Bagged_DOTM$summary,
  DSTM = results_list$Bagged_DSTM$summary,
  OTM = results_list$Bagged_OTM$summary,
  STM = results_list$Bagged_STM$summary
)
summaries
```

```
##         DOTM      DSTM      OTM       STM
## mean    0.9001437 0.858477  0.8663793 0.8570402
## median  1         1         1         1
## sd      0.2077699 0.2553545 0.2432567 0.2534815
## min     0         0         0         0
```

```
## max    1          1          1          1
```

2 ACD

2.1 Frequência Anual

```r
# Process the model type and calculate coverage rates
process_model <- function(model_key, label, M3, results, n_rows = 3003, n_cols = 18) {
  lo <- matrix(NA, nrow = n_rows, ncol = n_cols)
  hi <- matrix(NA, nrow = n_rows, ncol = n_cols)
  obs <- matrix(NA, nrow = n_rows, ncol = n_cols)
  acd_vec <- c()
  c <- 1

  for (i in 1:n_rows) {
    x <- M3[[i]]$x
    h <- M3[[i]]$h
    if (M3[[i]]$period == "YEARLY") {
      obs[i, 1:h] <- M3[[i]]$xx
      lo[i, 1:h] <- results[[model_key]][[c]]$lower[, 3] # 95% interval
      hi[i, 1:h] <- results[[model_key]][[c]]$upper[, 3]
      c <- c + 1
      acd_vec <- c(
        acd_vec,
        acd(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h], alpha = 0.05)
      )
    }
  }

  return(mean(acd_vec))
}

# Process each model and store the results
results_list <- list(
  STM = process_model("stm_anua", "STM", M3, results),
  OTM = process_model("otm_anua", "OTM", M3, results),
  DSTM = process_model("dstm_anua", "DSTM", M3, results),
  DOTM = process_model("dotm_anua", "DOTM", M3, results)
)

# ACD medio
results_list
```

```
## $STM
## [1] 0.2631266
##
## $OTM
## [1] 0.2682946
##
## $DSTM
## [1] 0.2628165
##
## $DOTM
## [1] 0.2466925
```

```r
results_list <- list(
  Bagged_STM = process_model("bagged_stm_anua", "Bagged_STM", M3, results),
  Bagged_OTM = process_model("bagged_otm_anua", "Bagged_OTM", M3, results),
  Bagged_DSTM = process_model("bagged_dstm_anua", "Bagged_DSTM", M3, results),
  Bagged_DOTM = process_model("bagged_dotm_anua", "Bagged_DOTM", M3, results)
)

# ACD medio
results_list
```

```
## $Bagged_STM
## [1] 0.2691214
##
## $Bagged_OTM
## [1] 0.2599225
##
## $Bagged_DSTM
## [1] 0.2702584
##
## $Bagged_DOTM
## [1] 0.2544444
```

2.2 Frequência Mensal

```r
# Process the model type and calculate coverage rates
process_model <- function(model_key, label, M3, results, n_rows = 3003, n_cols = 18) {
  lo <- matrix(NA, nrow = n_rows, ncol = n_cols)
  hi <- matrix(NA, nrow = n_rows, ncol = n_cols)
  obs <- matrix(NA, nrow = n_rows, ncol = n_cols)
  acd_vec <- c()
  c <- 1

  for (i in 1:n_rows) {
    x <- M3[[i]]$x
    h <- M3[[i]]$h
    if (M3[[i]]$period == "MONTHLY") {
      obs[i, 1:h] <- M3[[i]]$xx
      lo[i, 1:h] <- results[[model_key]][[c]]$lower[, 3] # 95% interval
      hi[i, 1:h] <- results[[model_key]][[c]]$upper[, 3]
      c <- c + 1
      acd_vec <- c(
        acd_vec,
        acd(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
      )
    }
  }
  return(mean(acd_vec))
}

# Process each model and store the results
results_list <- list(
  STM = process_model("stm_mens", "STM", M3, results),
  OTM = process_model("otm_mens", "OTM", M3, results),
```

```r
  DSTM = process_model("dstm_mens", "DSTM", M3, results),
  DOTM = process_model("dotm_mens", "DOTM", M3, results)
)

results_list
```

```
## $STM
## [1] 0.1257937
##
## $OTM
## [1] 0.1231793
##
## $DSTM
## [1] 0.1279256
##
## $DOTM
## [1] 0.1227202
```

```r
results_list <- list(
  Bagged_STM = process_model("bagged_stm_mens", "Bagged_STM", M3, results),
  Bagged_OTM = process_model("bagged_otm_mens", "Bagged_OTM", M3, results),
  Bagged_DSTM = process_model("bagged_dstm_mens", "Bagged_DSTM", M3, results),
  Bagged_DOTM = process_model("bagged_dotm_mens", "Bagged_DOTM", M3, results)
)


# ACD medio
results_list
```

```
## $Bagged_STM
## [1] 0.1222222
##
## $Bagged_OTM
## [1] 0.1165422
##
## $Bagged_DSTM
## [1] 0.125249
##
## $Bagged_DOTM
## [1] 0.1182073
```

2.3 Frequência Trimestral

```r
# Process the model type and calculate coverage rates
process_model <- function(model_key, label, M3, results, n_rows = 3003, n_cols = 18) {
  lo <- matrix(NA, nrow = n_rows, ncol = n_cols)
  hi <- matrix(NA, nrow = n_rows, ncol = n_cols)
  obs <- matrix(NA, nrow = n_rows, ncol = n_cols)
  acd_vec <- c()
  c <- 1

  for (i in 1:n_rows) {
    x <- M3[[i]]$x
    h <- M3[[i]]$h
```

```r
    if (M3[[i]]$period == "QUARTERLY") {
      obs[i, 1:h] <- M3[[i]]$xx
      lo[i, 1:h] <- results[[model_key]][[c]]$lower[, 3] # 95% interval
      hi[i, 1:h] <- results[[model_key]][[c]]$upper[, 3]
      c <- c + 1
      acd_vec <- c(
        acd_vec,
        acd(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
      )
    }
  }
  return(mean(acd_vec))
}

# Process each model and store the results
results_list <- list(
  STM = process_model("stm_trim", "STM", M3, results),
  OTM = process_model("otm_trim", "OTM", M3, results),
  DSTM = process_model("dstm_trim", "DSTM", M3, results),
  DOTM = process_model("dotm_trim", "DOTM", M3, results)
)

# ACD medio
results_list
```

```
## $STM
## [1] 0.1888558
##
## $OTM
## [1] 0.1914683
##
## $DSTM
## [1] 0.1825728
##
## $DOTM
## [1] 0.1918981
```

```r
results_list <- list(
  Bagged_STM = process_model("bagged_stm_trim", "Bagged_STM", M3, results),
  Bagged_OTM = process_model("bagged_otm_trim", "Bagged_OTM", M3, results),
  Bagged_DSTM = process_model("bagged_dstm_trim", "Bagged_DSTM", M3, results),
  Bagged_DOTM = process_model("bagged_dotm_trim", "Bagged_DOTM", M3, results)
)


# ACD medio
results_list
```

```
## $Bagged_STM
## [1] 0.2005291
##
## $Bagged_OTM
## [1] 0.1937831
##
```

```
## $Bagged_DSTM
## [1] 0.1945767
##
## $Bagged_DOTM
## [1] 0.2019511
```

2.4 Demais Frequências

```r
# Process the model type and calculate coverage rates
process_model <- function(model_key, label, M3, results, n_rows = 3003, n_cols = 18) {
  lo <- matrix(NA, nrow = n_rows, ncol = n_cols)
  hi <- matrix(NA, nrow = n_rows, ncol = n_cols)
  obs <- matrix(NA, nrow = n_rows, ncol = n_cols)
  acd_vec <- c()
  c <- 1

  for (i in 1:n_rows) {
    x <- M3[[i]]$x
    h <- M3[[i]]$h
    if (M3[[i]]$period == "OTHER") {
      obs[i, 1:h] <- M3[[i]]$xx
      lo[i, 1:h] <- results[[model_key]][[c]]$lower[, 3] # 95% interval
      hi[i, 1:h] <- results[[model_key]][[c]]$upper[, 3]
      c <- c + 1
      acd_vec <- c(
        acd_vec,
        acd(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
      )
    }
  }
  return(mean(acd_vec))
}

# Process each model and store the results
results_list <- list(
  STM = process_model("stm_outr", "STM", M3, results),
  OTM = process_model("otm_outr", "OTM", M3, results),
  DSTM = process_model("dstm_outr", "DSTM", M3, results),
  DOTM = process_model("dotm_outr", "DOTM", M3, results)
)

# ACD medio
results_list
```

```
## $STM
## [1] 0.1316092
##
## $OTM
## [1] 0.1340517
##
## $DSTM
## [1] 0.1306034
##
## $DOTM
```

```
## [1] 0.09137931
```

```r
results_list <- list(
  Bagged_STM = process_model("bagged_stm_outr", "Bagged_STM", M3, results),
  Bagged_OTM = process_model("bagged_otm_outr", "Bagged_OTM", M3, results),
  Bagged_DSTM = process_model("bagged_dstm_outr", "Bagged_DSTM", M3, results),
  Bagged_DOTM = process_model("bagged_dotm_outr", "Bagged_DOTM", M3, results)
)


# ACD medio
results_list
```

```
## $Bagged_STM
## [1] 0.1579023
##
## $Bagged_OTM
## [1] 0.1502874
##
## $Bagged_DSTM
## [1] 0.1587644
##
## $Bagged_DOTM
## [1] 0.1222701
```

3 MSIS

3.1 Frequência Anual

```r
# Process the model type and calculate coverage rates
process_model <- function(model_key, label, M3, results, n_rows = 3003, n_cols = 18) {
  lo <- matrix(NA, nrow = n_rows, ncol = n_cols)
  hi <- matrix(NA, nrow = n_rows, ncol = n_cols)
  obs <- matrix(NA, nrow = n_rows, ncol = n_cols)
  msis_vec <- c()
  c <- 1

  for (i in 1:n_rows) {
    x <- M3[[i]]$x
    h <- M3[[i]]$h
    if (M3[[i]]$period == "YEARLY") {
      snaive_forecast <- snaive(x, h = h)$mean
      obs[i, 1:h] <- M3[[i]]$xx
      scale = mean(abs(snaive_forecast - obs[i, 1:h]))
      lo[i, 1:h] <- results[[model_key]][[c]]$lower[, 3] # 95% interval
      hi[i, 1:h] <- results[[model_key]][[c]]$upper[, 3]
      c <- c + 1
      msis_vec <- c(
        msis_vec,
        msis(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h], scale = scale, alpha=0.05)
      )
    }
  }
```

```r
  return(mean(msis_vec))
}

# Process each model and store the results
results_list <- list(
  STM = process_model("stm_anua", "STM", M3, results),
  OTM = process_model("otm_anua", "OTM", M3, results),
  DSTM = process_model("dstm_anua", "DSTM", M3, results),
  DOTM = process_model("dotm_anua", "DOTM", M3, results)
)

# MSIS medio
results_list
```

```
## $STM
## [1] 9.660514
##
## $OTM
## [1] 10.46774
##
## $DSTM
## [1] 9.792607
##
## $DOTM
## [1] 10.42804
```

```r
results_list <- list(
  Bagged_STM = process_model("bagged_stm_anua", "Bagged_STM", M3, results),
  Bagged_OTM = process_model("bagged_otm_anua", "Bagged_OTM", M3, results),
  Bagged_DSTM = process_model("bagged_dstm_anua", "Bagged_DSTM", M3, results),
  Bagged_DOTM = process_model("bagged_dotm_anua", "Bagged_DOTM", M3, results)
)

# MSIS medio
results_list
```

```
## $Bagged_STM
## [1] 9.80371
##
## $Bagged_OTM
## [1] 10.167
##
## $Bagged_DSTM
## [1] 9.826352
##
## $Bagged_DOTM
## [1] 10.57899
```

3.2 Frequência Mensal

```r
# Process the model type and calculate coverage rates
process_model <- function(model_key, label, M3, results, n_rows = 3003, n_cols = 18) {
  lo <- matrix(NA, nrow = n_rows, ncol = n_cols)
  hi <- matrix(NA, nrow = n_rows, ncol = n_cols)
```

```r
  obs <- matrix(NA, nrow = n_rows, ncol = n_cols)
  msis_vec <- c()
  c <- 1

  for (i in 1:n_rows) {
    x <- M3[[i]]$x
    h <- M3[[i]]$h
    if (M3[[i]]$period == "MONTHLY") {
      snaive_forecast <- snaive(x, h = h)$mean
      obs[i, 1:h] <- M3[[i]]$xx
      scale = mean(abs(snaive_forecast - obs[i, 1:h]))
      lo[i, 1:h] <- results[[model_key]][[c]]$lower[, 3] # 95% interval
      hi[i, 1:h] <- results[[model_key]][[c]]$upper[, 3]
      c <- c + 1
      msis_vec <- c(
        msis_vec,
        msis(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h], scale = scale, alpha=0.05)
      )
    }
  }
  return(mean(msis_vec))
}

# Process each model and store the results
results_list <- list(
  STM = process_model("stm_mens", "STM", M3, results),
  OTM = process_model("otm_mens", "OTM", M3, results),
  DSTM = process_model("dstm_mens", "DSTM", M3, results),
  DOTM = process_model("dotm_mens", "DOTM", M3, results)
)

results_list
```

```
## $STM
## [1] 6.552753
##
## $OTM
## [1] 6.560777
##
## $DSTM
## [1] 6.622396
##
## $DOTM
## [1] 6.658979
```

```r
results_list <- list(
  Bagged_STM = process_model("bagged_stm_mens", "Bagged_STM", M3, results),
  Bagged_OTM = process_model("bagged_otm_mens", "Bagged_OTM", M3, results),
  Bagged_DSTM = process_model("bagged_dstm_mens", "Bagged_DSTM", M3, results),
  Bagged_DOTM = process_model("bagged_dotm_mens", "Bagged_DOTM", M3, results)
)


# MSIS medio
```

```r
results_list
```

```
## $Bagged_STM
## [1] 6.60054
##
## $Bagged_OTM
## [1] 6.579045
##
## $Bagged_DSTM
## [1] 6.683228
##
## $Bagged_DOTM
## [1] 6.663092
```

3.3 Frequência Trimestral

```r
# Process the model type and calculate coverage rates
process_model <- function(model_key, label, M3, results, n_rows = 3003, n_cols = 18) {
  lo <- matrix(NA, nrow = n_rows, ncol = n_cols)
  hi <- matrix(NA, nrow = n_rows, ncol = n_cols)
  obs <- matrix(NA, nrow = n_rows, ncol = n_cols)
  msis_vec <- c()
  c <- 1

  for (i in 1:n_rows) {
    x <- M3[[i]]$x
    h <- M3[[i]]$h
    if (M3[[i]]$period == "QUARTERLY") {
      snaive_forecast <- snaive(x, h = h)$mean
      obs[i, 1:h] <- M3[[i]]$xx
      scale = mean(abs(snaive_forecast - obs[i, 1:h]))
      lo[i, 1:h] <- results[[model_key]][[c]]$lower[, 3] # 95% interval
      hi[i, 1:h] <- results[[model_key]][[c]]$upper[, 3]
      c <- c + 1
      msis_vec <- c(
        msis_vec,
        msis(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h], scale = scale, alpha=0.05)
      )
    }
  }
  return(mean(msis_vec))
}

# Process each model and store the results
results_list <- list(
  STM = process_model("stm_trim", "STM", M3, results),
  OTM = process_model("otm_trim", "OTM", M3, results),
  DSTM = process_model("dstm_trim", "DSTM", M3, results),
  DOTM = process_model("dotm_trim", "DOTM", M3, results)
)

# MSIS medio
results_list
```

```
## $STM
## [1] 7.429898
##
## $OTM
## [1] 7.750784
##
## $DSTM
## [1] 7.388885
##
## $DOTM
## [1] 8.421939
```

```
results_list <- list(
  Bagged_STM = process_model("bagged_stm_trim", "Bagged_STM", M3, results),
  Bagged_OTM = process_model("bagged_otm_trim", "Bagged_OTM", M3, results),
  Bagged_DSTM = process_model("bagged_dstm_trim", "Bagged_DSTM", M3, results),
  Bagged_DOTM = process_model("bagged_dotm_trim", "Bagged_DOTM", M3, results)
)


# MSIS medio
results_list
```

```
## $Bagged_STM
## [1] 7.32047
##
## $Bagged_OTM
## [1] 7.564455
##
## $Bagged_DSTM
## [1] 7.29123
##
## $Bagged_DOTM
## [1] 8.443794
```

3.4 Demais Frequências

```
# Process the model type and calculate coverage rates
process_model <- function(model_key, label, M3, results, n_rows = 3003, n_cols = 18) {
  lo <- matrix(NA, nrow = n_rows, ncol = n_cols)
  hi <- matrix(NA, nrow = n_rows, ncol = n_cols)
  obs <- matrix(NA, nrow = n_rows, ncol = n_cols)
  msis_vec <- c()
  c <- 1

  for (i in 1:n_rows) {
    x <- M3[[i]]$x
    h <- M3[[i]]$h
    if (M3[[i]]$period == "OTHER") {
      snaive_forecast <- snaive(x, h = h)$mean
      obs[i, 1:h] <- M3[[i]]$xx
      scale = mean(abs(snaive_forecast - obs[i, 1:h]))
      lo[i, 1:h] <- results[[model_key]][[c]]$lower[, 3] # 95% interval
      hi[i, 1:h] <- results[[model_key]][[c]]$upper[, 3]
```

```r
      c <- c + 1
      msis_vec <- c(
        msis_vec,
        msis(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h], scale = scale, alpha=0.05)
      )
    }
  }
  return(mean(msis_vec))
}

# Process each model and store the results
results_list <- list(
  STM = process_model("stm_outr", "STM", M3, results),
  OTM = process_model("otm_outr", "OTM", M3, results),
  DSTM = process_model("dstm_outr", "DSTM", M3, results),
  DOTM = process_model("dotm_outr", "DOTM", M3, results)
)

# MSIS medio
results_list
```

```
## $STM
## [1] 6.663108
##
## $OTM
## [1] 6.668059
##
## $DSTM
## [1] 6.733046
##
## $DOTM
## [1] 6.334265
```

```r
results_list <- list(
  Bagged_STM = process_model("bagged_stm_outr", "Bagged_STM", M3, results),
  Bagged_OTM = process_model("bagged_otm_outr", "Bagged_OTM", M3, results),
  Bagged_DSTM = process_model("bagged_dstm_outr", "Bagged_DSTM", M3, results),
  Bagged_DOTM = process_model("bagged_dotm_outr", "Bagged_DOTM", M3, results)
)


# MSIS medio
results_list
```

```
## $Bagged_STM
## [1] 6.719366
##
## $Bagged_OTM
## [1] 6.662957
##
## $Bagged_DSTM
## [1] 6.756272
##
## $Bagged_DOTM
## [1] 6.476561
```